

MICRONYX PS02

V2 R&D PHASE-WISE WORK PLAN

Minimum expectations • Strong targets • Maximum research extensions • Deliverables • Completion criteria

Purpose: Turn the frozen MICRONYX v1 prototype into a technically rigorous, experimentally validated R&D system without adding complexity unless it produces measurable improvement.

MICRONYX V2 — SYSTEM OBJECTIVE

Develop MICRONYX as an end-to-end, sensorless AI navigation-error recovery system for semiconductor wafer inspection: robustly recover target coordinates from differently sampled search/reference observations, resolve repetitive-structure ambiguity, quantify uncertainty, and convert the recovered position into a simulated navigation correction.

CORE PIPELINE

Search + Reference → Acquisition normalization → Candidate retrieval → Multi-scale representation → Hard-negative-aware ranking → Structural/geometric verification → Sub-pixel refinement → Confidence + uncertainty → (x,y) → Navigation correction $\Delta x, \Delta y$ → Recovery/re-acquisition simulation

PHASE 0 — V1 FREEZE & BASELINE

Objective: Create the immutable scientific baseline.

MINIMUM — mandatory

- Record V1 model/version, dataset/version, candidate count, recall, Top-1/Top-5, localization error, runtime and failure cases.
- Preserve the v1.0-submission Git tag.

TARGET — strong expected outcome

- Create v1_baseline.json/csv and an automated evaluator.

MAXIMUM — research extension

- Automated reproducibility script and baseline failure taxonomy.

DELIVERABLES

baseline/metrics.csv; baseline/metrics.json; baseline/failure_cases/; baseline/README.md

DONE WHEN: Done when V1 metrics are reproducible and V2 improvement can be measured exactly.

PHASE 1 — DATASET V2 & BENCHMARK INFRASTRUCTURE

Objective: Make experiments scientifically trustworthy.

MINIMUM — mandatory

- Support periodic, quasiperiodic and aperiodic structures; DRAM and FinFET; noise, blur, contrast, intensity and translation variation; strict train/validation/test splits.
- Freeze the test set.

TARGET — strong expected outcome

- Add pitch, line width, spacing, orientation, scale and structural-density controls plus metadata for every sample.

MAXIMUM — research extension

- Configurable domain randomization and easy/medium/hard difficulty groups.

DELIVERABLES

dataset_v2/generator/; train/; validation/; test/; metadata/; benchmark_v2/evaluate.py

DONE WHEN: Done when one evaluation command reports recall, Top-1/5/10, error statistics and runtime automatically.

PHASE 2 — CANDIDATE GENERATION

Objective: Ensure the correct target enters the candidate pool.

MINIMUM — mandatory

- Measure Recall @ 1/3/5/10 px for K=25/50/100/250.
- Analyze candidate count versus recall and runtime.

TARGET — strong expected outcome

- Improve proposal quality while controlling candidate count.

MAXIMUM — research extension

- Combine DoG, correlation, gradient and multi-scale proposals with clustering/NMS if justified.

DELIVERABLES

`candidate_generation_v2.py`; `candidate_recall_results.csv`; `candidate_recall_report.pdf`; `visualizations/`

DONE WHEN: Done when the team can distinguish candidate-generation failure from ranking failure.

PHASE 3 — HARD-NEGATIVE MINING

Objective: Attack visually similar wrong candidates.

MINIMUM — mandatory

- Collect high-scoring incorrect candidates and build a hard-negative dataset.
- Retrain XGBoost.

TARGET — strong expected outcome

- Compare V1 XGBoost versus hard-negative XGBoost on the same frozen test set.

MAXIMUM — research extension

- Iterative train → infer → mine → retrain loop until improvement saturates.

DELIVERABLES

`hard_negative_mining/`; `hard_negatives.csv`; `training_results.csv`; `model/`; `report.md`

DONE WHEN: Done when ranking improves without degrading candidate recall. If improvement is negligible, stop.

PHASE 4 — MULTI-SCALE REPRESENTATION

Objective: Use local appearance plus surrounding structure.

MINIMUM — mandatory

- Extract local, medium and large context representations.
- Evaluate useful intensity, gradient, edge, orientation, morphology, texture, frequency/periodicity and spatial-context features.

TARGET — strong expected outcome

- Compare V1 features versus multi-scale features and run ablations.

MAXIMUM — research extension

- Investigate learned embeddings only if engineered features hit a measurable ceiling.

DELIVERABLES

`multiscale/feature_extractor.py`; `feature_ablation.csv`; `importance.json`; `report.md`

DONE WHEN: Done when evidence shows whether context improves discrimination.

PHASE 5 — ADVANCED RANKING

Objective: Strengthen ranking while retaining reproducibility.

MINIMUM — mandatory

- Optimize current XGBoost: feature selection, hyperparameters, balance and hard-negative weighting.

TARGET — strong expected outcome

- Compare suitable pointwise, pairwise and listwise ranking approaches.

MAXIMUM — research extension

- Evaluate a learned matching/ranking architecture only if it fairly beats the simpler baseline.

DELIVERABLES

ranking_v2/models/; training/; results.csv; comparison.csv

DONE WHEN: Done when the chosen ranking method is quantitatively justified.

PHASE 6 — STRUCTURAL / GEOMETRIC VERIFICATION

Objective: Independently verify top-ranked candidates.

MINIMUM — mandatory

- Compare reference/candidate gradients, orientation, edge layout, spatial arrangement and multi-scale context.

TARGET — strong expected outcome

- Combine ML and structural scores and rerank.

MAXIMUM — research extension

- Investigate robust correspondence/transformation methods if justified by failure data.

DELIVERABLES

geometry/verifier.py; scores.csv; visualizations/; report.md

DONE WHEN: Done when verification measurably reduces false positives on difficult repetitive structures.

PHASE 7 — SUB-PIXEL REFINEMENT

Objective: Move beyond integer-pixel localization.

MINIMUM — mandatory

- Inspect the response surface around the winning candidate.

TARGET — strong expected outcome

- Implement a robust peak refinement method such as quadratic/Gaussian fitting and compare integer versus refined error.

MAXIMUM — research extension

- Compare multiple refinement methods if useful.

DELIVERABLES

refinement/refine.py; comparison.csv; report.md

DONE WHEN: Done when refinement measurably improves localization accuracy.

PHASE 8 — CONFIDENCE & UNCERTAINTY

Objective: Make confidence scientifically defensible.

MINIMUM — mandatory

- Separate raw model score from calibrated confidence.

TARGET — strong expected outcome

- Perform reliability analysis and calibration-error measurement; evaluate Platt or isotonic calibration where appropriate.

MAXIMUM — research extension

- Estimate coordinate uncertainty and high/medium/low confidence states.

DELIVERABLES

uncertainty/calibration.py; calibration_results.csv; reliability_plot.png; report.md

DONE WHEN: Done when the meaning and calibration quality of displayed confidence can be defended.

PHASE 9 — NAVIGATION ERROR RECOVERY

Objective: Connect localization to the industrial navigation problem.

MINIMUM — mandatory

- Simulate true target → navigation offset → observed position → MICRONYX prediction → correction. Measure initial, recovered and residual error.

TARGET — strong expected outcome

- Build a virtual stage with acquisition, prediction, correction and re-acquisition.

MAXIMUM — research extension

- Implement closed-loop recovery and measure convergence, stability and failure rate.

DELIVERABLES

`navigation/simulator.py; recovery_results.csv; convergence_plots/; report.md`

DONE WHEN: Done when recovered coordinates produce a measurable simulated navigation correction.

PHASE 10 — ROBUSTNESS & STRESS TESTING

Objective: Find the system's failure envelope.

MINIMUM — mandatory

- Test noise, blur, contrast, translation, scale, periodicity, quasiperiodicity and aperiodicity.

TARGET — strong expected outcome

- Create low/medium/high perturbation matrices and heatmaps.

MAXIMUM — research extension

- Test combined perturbations and difficult structural cases.

DELIVERABLES

`stress_tests/runner.py; robustness_results.csv; heatmaps/; report.md`

DONE WHEN: Done when the team knows where and why the system fails.

PHASE 11 — BASELINE COMPARISON

Objective: Prove MICRONYX adds measurable value.

MINIMUM — mandatory

- Compare Template Matching, DoG, current XGBoost and MICRONYX V2.

TARGET — strong expected outcome

- Add handcrafted scoring, hard-negative XGBoost and multi-scale ranking where appropriate.

MAXIMUM — research extension

- Add one competitive external/reference method if fairly implementable.

DELIVERABLES

`benchmark_comparison.csv`

DONE WHEN: Done when every method is evaluated under identical conditions using recall, Top-1/5, error and runtime.

PHASE 12 — ABLATION STUDY

Objective: Prove each major component contributes.

MINIMUM — mandatory

- Run baseline, +hard negatives, +multi-scale, +geometry, +refinement and +calibration.

TARGET — strong expected outcome

- Run full removal tests and create a master ablation table.

MAXIMUM — research extension

- Add statistical analysis where justified.

DELIVERABLES

ablation/ablation_results.csv; plots/; report.md

DONE WHEN: Done when every major module has an evidence-based justification.

PHASE 13 — REAL SEM DOMAIN STUDY

Objective: Investigate synthetic-to-real transfer without corrupting the official benchmark.

MINIMUM — mandatory

- Acquire suitable public SEM data and analyze morphology, feature distributions, noise and contrast.

TARGET — strong expected outcome

- Study representation transfer and real-SEM failure patterns.

MAXIMUM — research extension

- Perform controlled domain adaptation as a separate research experiment.

DELIVERABLES

real_sem/dataset_notes/; domain_gap_analysis.csv; visualizations/; report.md

DONE WHEN: Done when the team can honestly state what transfers to real SEM and what does not. Do not claim official PS02 localization accuracy without suitable ground truth.

PHASE 14 — REPRODUCIBILITY

Objective: Make MICRONYX independently runnable.

MINIMUM — mandatory

- Provide one command such as `python evaluate.py`.

TARGET — strong expected outcome

- Automatically load model/test data, infer, calculate metrics and write a report.

MAXIMUM — research extension

- Test in a clean environment and document Python, dependencies, hardware, runtime, versions and seeds.

DELIVERABLES

evaluate.py; requirements.txt; reproducibility.md

DONE WHEN: Done when a teammate can reproduce reported numbers without editing source files.

PHASE 15 — V2 INTEGRATION

Objective: Integrate only validated components.

MINIMUM — mandatory

- Connect acquisition, candidates, multi-scale representation, ranker, verification, refinement and uncertainty.

TARGET — strong expected outcome

- Show integrated V2 beats V1 on the frozen benchmark.

MAXIMUM — research extension

- Optimize latency, memory and parallelization without changing scientific behavior.

DELIVERABLES

V2 inference pipeline; integration tests; V2 model package

DONE WHEN: Done when all retained components work together and measurable improvement is demonstrated.

PHASE 16 — FINAL V2 VALIDATION

Objective: Treat the frozen test set as the final exam.

MINIMUM — mandatory

- Report candidate recall, Top-1/5, median error, 95th-percentile error and runtime.

TARGET — strong expected outcome

- Add memory, calibration and navigation-recovery results.

MAXIMUM — research extension

- Perform repeated-run consistency and failure analysis.

DELIVERABLES

final_v2_results.csv; final_v2_report.pdf; v1_vs_v2.csv

DONE WHEN: Done when V1 versus V2 is fully documented and the test set is no longer used for tuning.

PHASE 17 — PRODUCTION PROTOTYPE V2

Objective: Deploy validated V2.

MINIMUM — mandatory

- Update FastAPI and frontend integration.

TARGET — strong expected outcome

- Display coordinate, confidence, uncertainty, candidates, navigation correction and residual error where supported.

MAXIMUM — research extension

- Run latency/stability testing and optimization.

DELIVERABLES

Updated API; updated frontend; deployment documentation

DONE WHEN: Done when the deployed frontend executes the validated V2 backend end-to-end.

PHASE 18 — FINAL EVIDENCE PACKAGE

Objective: Convert experiments into judge-readable evidence.

MINIMUM — mandatory

- Generate V1 vs V2, benchmark, ablation, robustness, failure and runtime tables.

TARGET — strong expected outcome

- Add candidate/ranking visualizations, confidence plots and navigation-recovery plots.

MAXIMUM — research extension

- Make every headline number traceable to a saved experiment.

DELIVERABLES

evidence/benchmark_tables/; plots/; failure_cases/

DONE WHEN: Done when every important performance claim is traceable to reproducible evidence.

PHASE 19 — FINAL R&D REPORT

Objective: Document the entire research and engineering process.

MINIMUM — mandatory

- Problem, industrial context, V1 baseline, V2 methodology, dataset, algorithms, evaluation and results.

TARGET — strong expected outcome

- Include hard negatives, multi-scale, geometry, refinement, uncertainty, navigation simulation, robustness, baselines, ablations and failure analysis.

MAXIMUM — research extension

- Include real-SEM study, limitations and reproducibility with strict separation between measured results and future targets.

DELIVERABLES

Technical report PDF; methods appendix; reproducibility appendix

DONE WHEN: Done when a technical reader can understand, reproduce and critically evaluate the system.

PHASE 20 — FINAL MICRONYX V2

Objective: Deliver the complete research-grade system.

MINIMUM — mandatory

- Working prototype, V2 model, evaluation script, dataset generator, benchmark suite, trained model and documentation.

TARGET — strong expected outcome

- Complete end-to-end demonstration with measured V2 evidence.

MAXIMUM — research extension

- Production optimization and a research package suitable for further development/publication if results justify it.

DELIVERABLES

Final repository; deployed system; final report; final PPT; demo video; future scope

DONE WHEN: Done when the project is reproducible, measurable, deployable, explainable and defensible as an R&D system.

GLOBAL RULES FOR EVERY TEAMMATE

- Every phase follows: Hypothesis → Implementation → Experiment → Metric → Baseline comparison → Keep/Reject.
- Maximum scope is optional. Do not implement advanced extensions unless the minimum version produces a measurable result or exposes a justified limitation.
- Never tune the frozen test set. Use training/validation for development and preserve the test set for final evaluation.
- Do not add CNNs, transformers or other advanced models merely for novelty. A simpler method that wins the benchmark is the correct engineering choice.
- Do not claim 95% performance, calibrated confidence, real-SEM accuracy or industrial deployment capability until corresponding evidence exists.
- Keep official PS02 benchmark results separate from external real-SEM studies unless the external data has appropriate paired ground truth.
- Every final performance number in the PPT/report must trace to a saved experiment and reproducible evaluation.

FINAL SYSTEM IDENTITY

MICRONYX — Sensorless AI Navigation-Error Recovery for Semiconductor Wafer Inspection

The final system should demonstrate robust target retrieval, learned ranking, structural verification, precise localization, defensible uncertainty and conversion of the recovered position into a measurable navigation correction.